Building Science

Lecture 2: Climate impacts on Building orientation and energy use

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Today’s outline

- Climate and Weather
- Building for appropriate climates
- Building orientation

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Building Science

Course content

1. General introduction to Building Science
2. Climate impacts on Building orientation
3. Energy efficiency and Passive control strategies
4. Thermal performance – heat transfer principles
5. Thermal comfort
6. Thermal performance – Insulation and thermal bridges

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Climate and Weather

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Climate

Weather!

Climate

Describe your home town’s weather?

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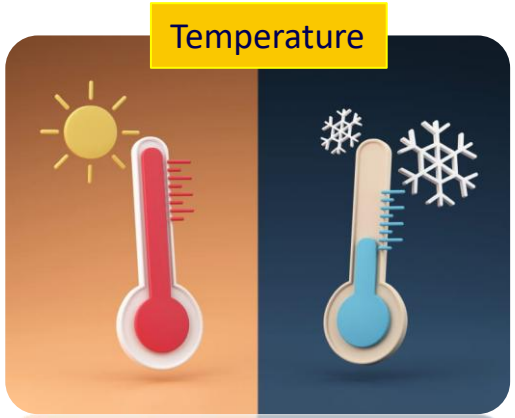
Climate

What are key features that help describe a city's CLIMATE?



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Climate

Other key features of a city's CLIMATE?

Windy

Cloud quantity

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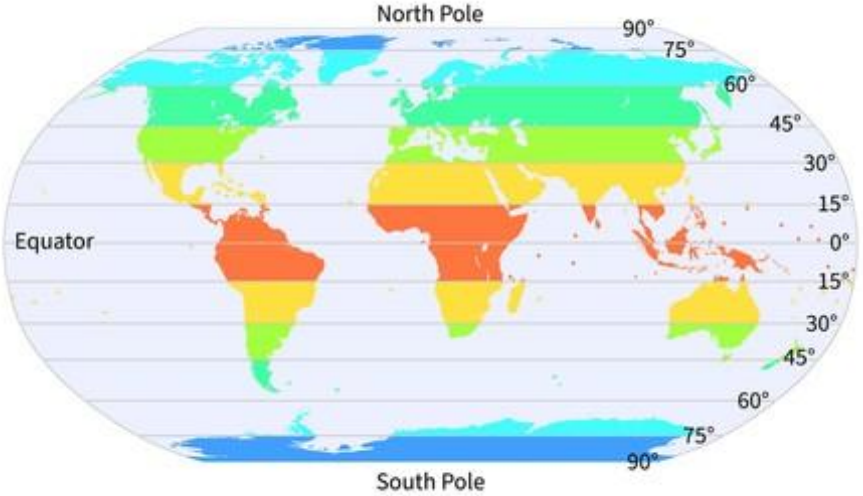
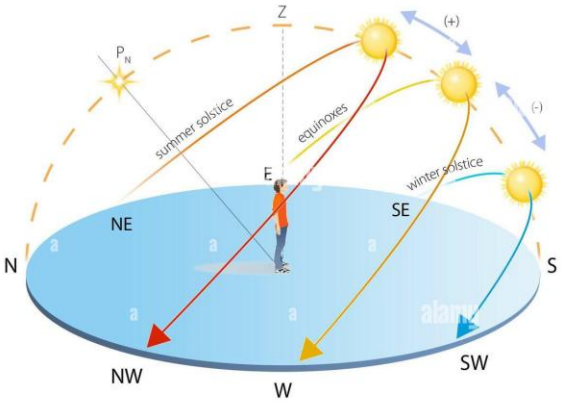
Climate

Solar access is related to *location* (on Earth) and *season*

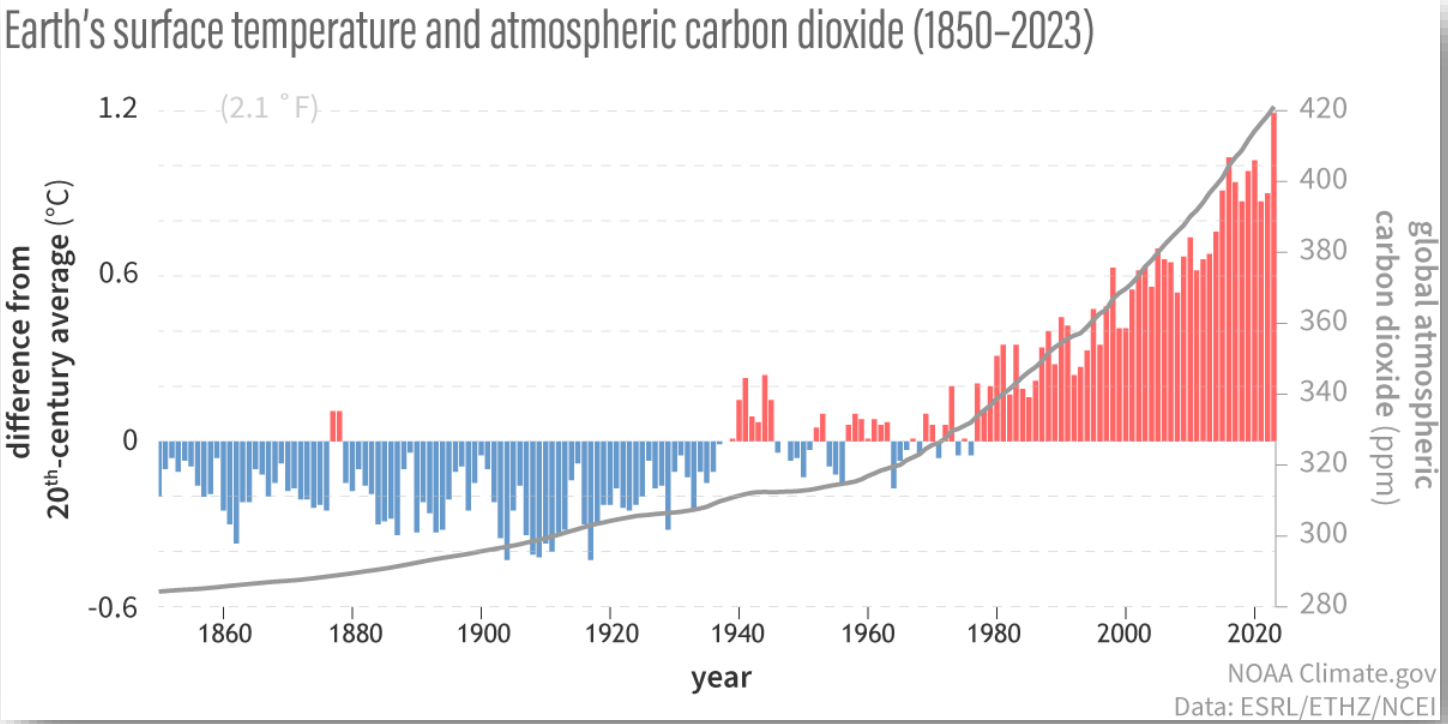


Latitude



Climate

Weather is becoming more unpredictable with increasing global warming



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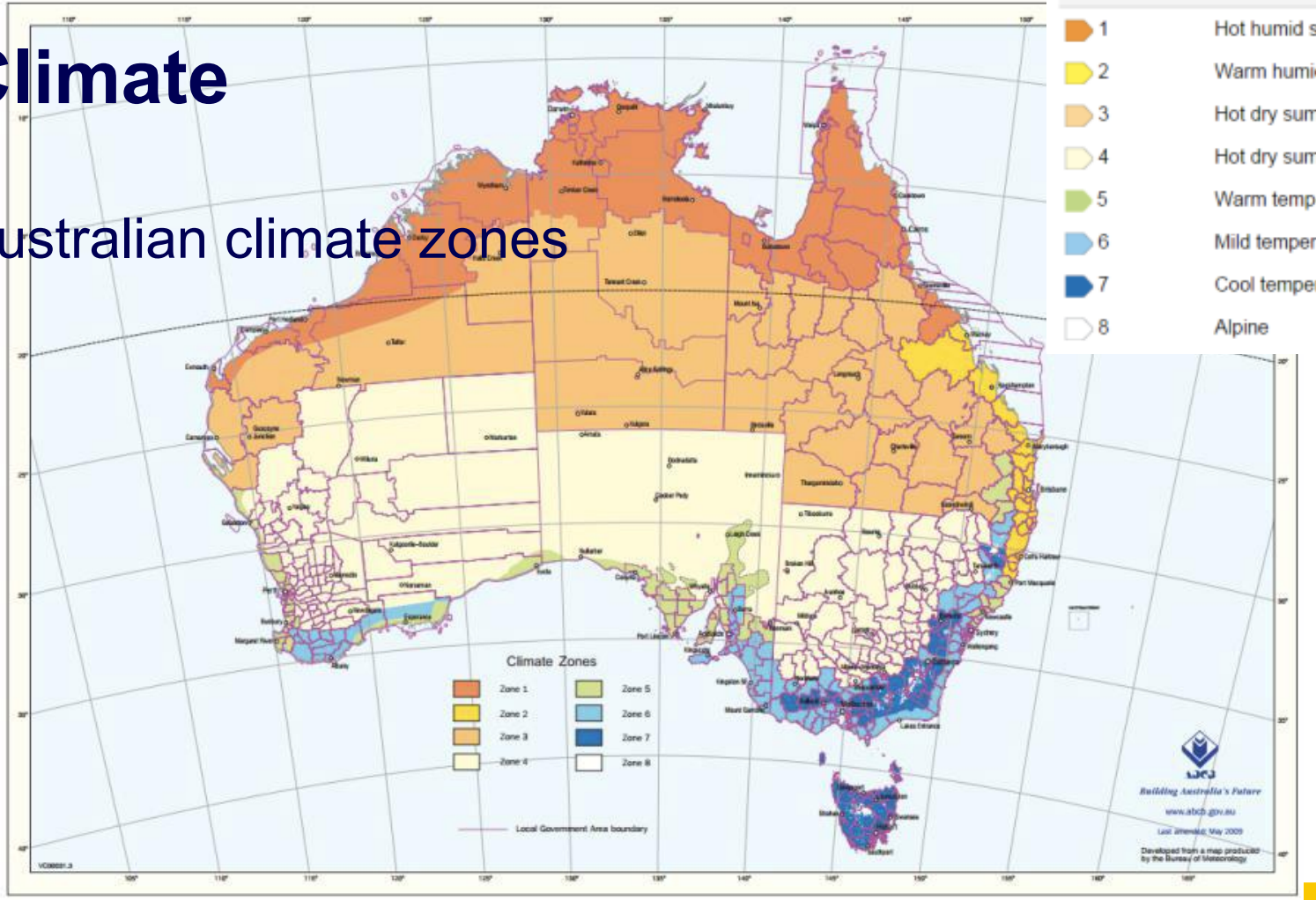
Climate

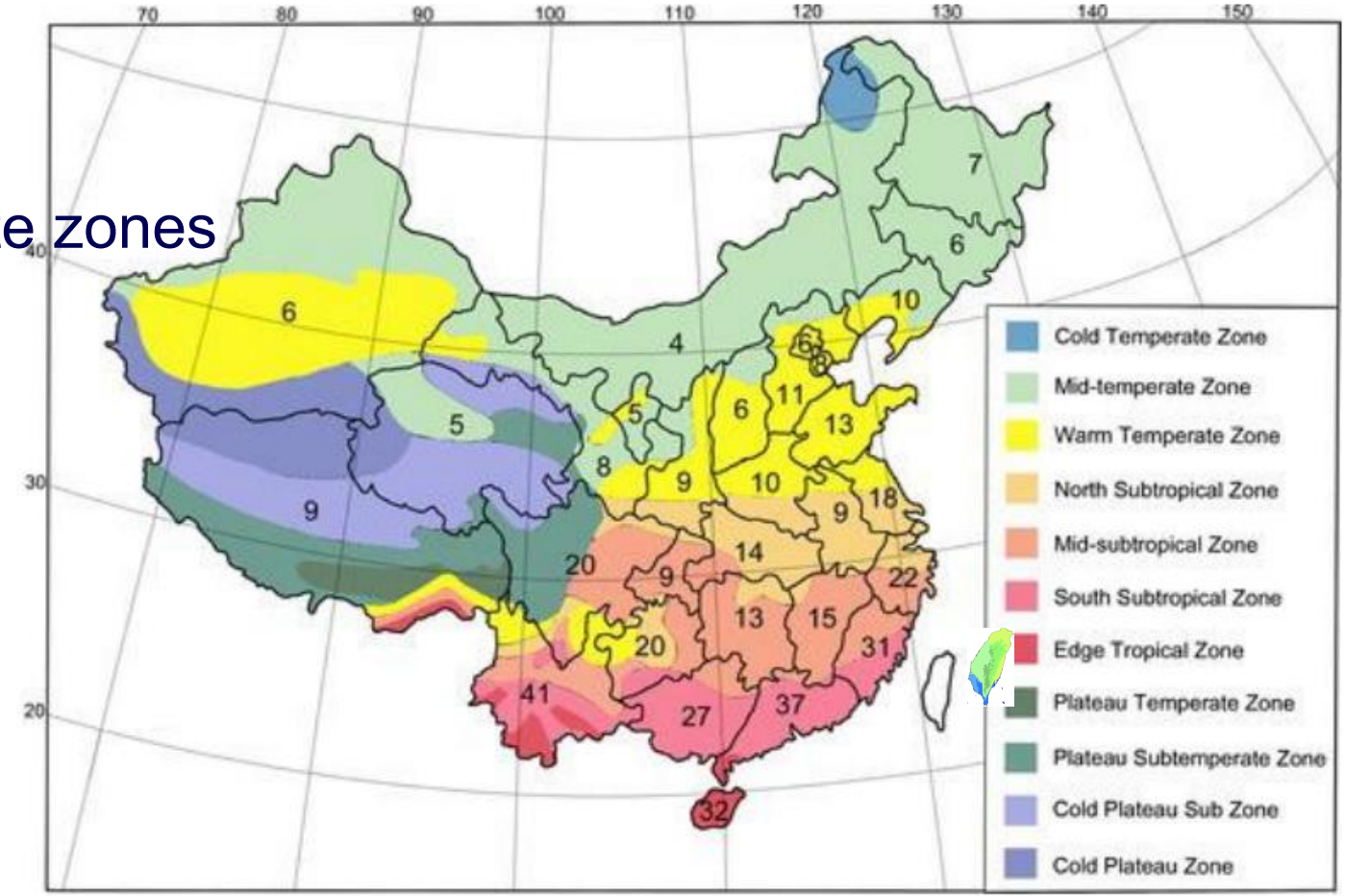
Australian climate zones



Zone	Description
1	Hot humid summer, warm winter
2	Warm humid summer, mild winter
3	Hot dry summer, warm winter
4	Hot dry summer, cool winter
5	Warm temperate
6	Mild temperate
7	Cool temperate
8	Alpine

Climate

China climate zones



Climate

- Climates for each city are used in the design of buildings. These are called '*Weather files*'

- Used in building design and modelling ***prior*** to building

- Formal scientific classification of weather for different locations are called the '*Koppen-Geiger Classification*'

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World Map of Köppen−Geiger Climate Classification

updated with CRU TS 2.1 temperature and VASClimO v1.1 precipitation data 1951 to 2000

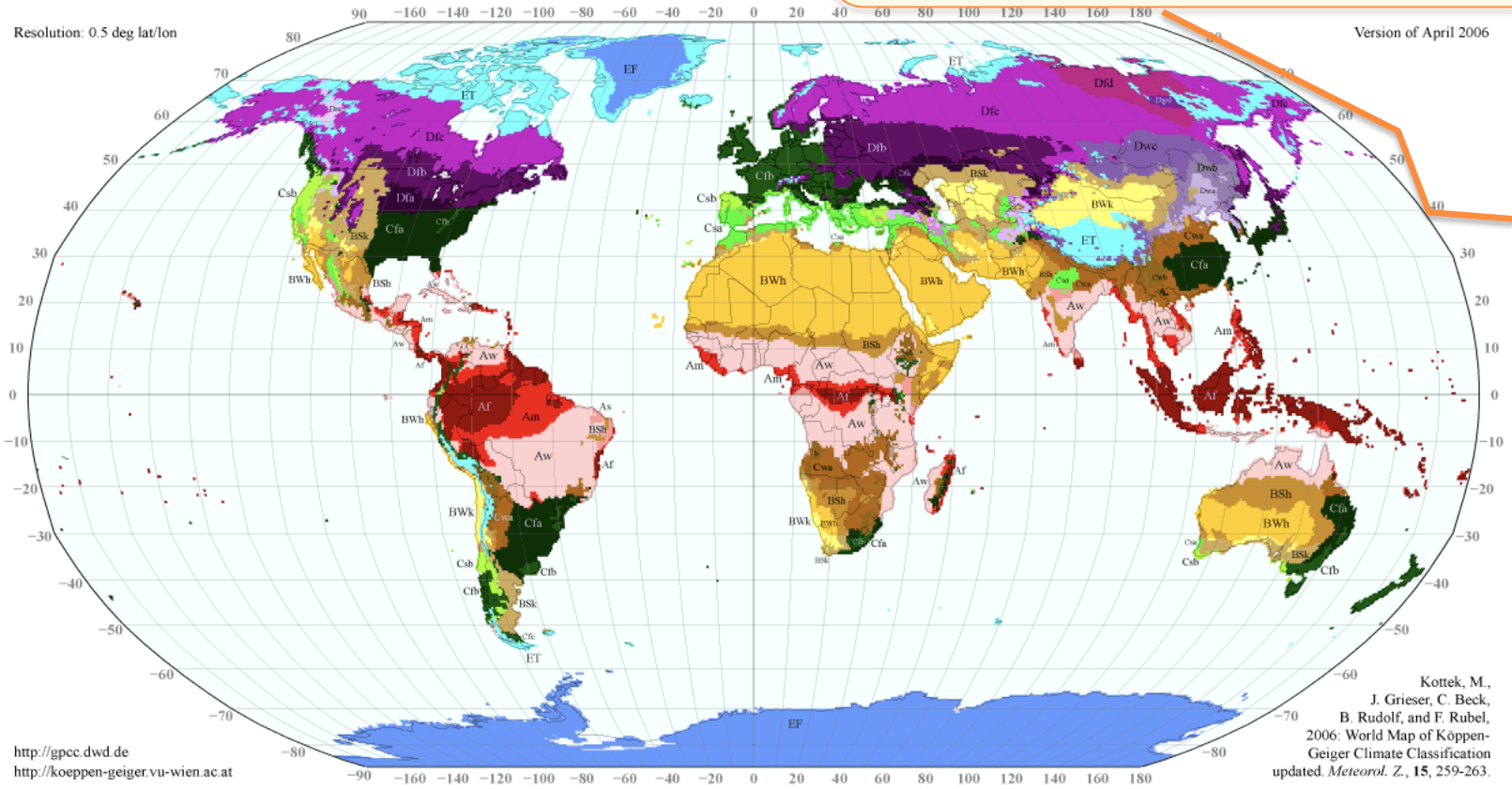
Af	Am	As	Aw	BWk	BWh	BSk	BSh	Cfa	Cfb	Cfc	Csa	Csb	Csc	Cwa
Cwb	Cwc	Dfa	Dfb	Dfc	Dfd	Dsa	Dsb	Dsc	Dsd	Dwa	Dwb	Dwc	Dwd	EF	ET

Main climates	Precipitation	Temperature	
A: equatorial	W: desert	h: hot arid	F: polar frost
B: arid	S: steppe	k: cold arid	T: polar tundra
C: warm temperate	f: fully humid	a: hot summer	
D: snow	s: summer dry	b: warm summer	
E: polar	w: winter dry	c: cool summer	
	m: monsoonal	d: extremely continental	

Important!

Resolution: 0.5 deg lat/lon

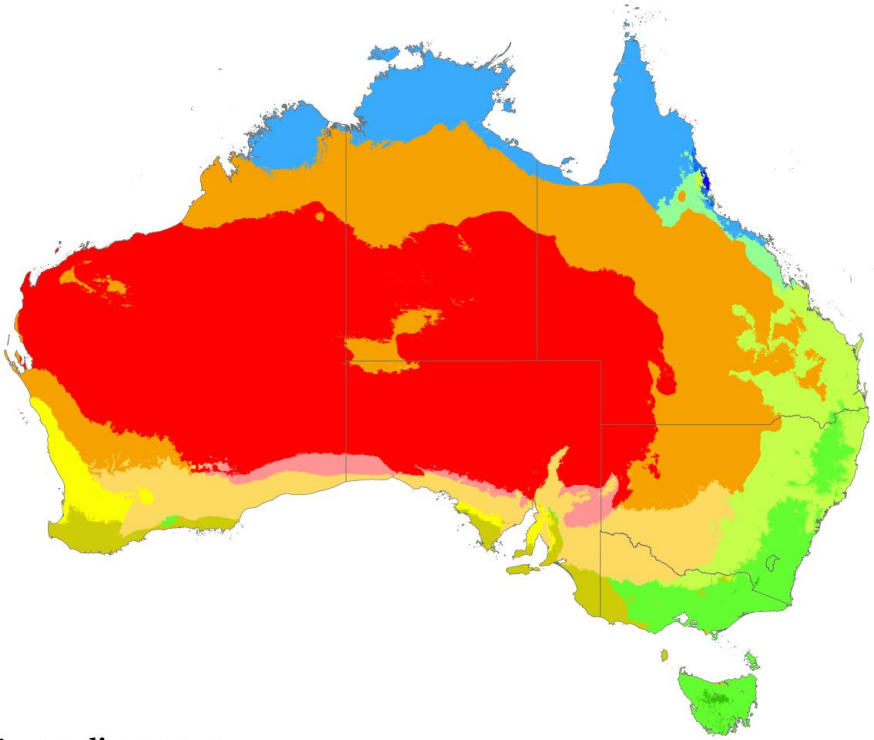
Version of April 2006



http://gpcc.dwd.de
http://koeppen-geiger.vu-wien.ac.at

Kottek, M., J. Grieser, C. Beck, B. Rudolf, and F. Rubel, 2006: World Map of Köppen-Geiger Climate Classification updated. *Meteorol. Z.*, **15**, 259-263.

Köppen climate types of Australia



Köppen climate type

- *ET (Tundra)*
- *Dfc (Subarctic)*
- *Cfc (Subpolar oceanic)*
- *Cfb (Oceanic)*
- *Cfa (Humid subtropical)*
- *Cwa (Humid subtropical)*
- *Csb (Warm-summer mediterranean)*
- *Csa (Hot-summer mediterranean)*
- *BSk (Cold semi-arid)*
- *BSh (Hot semi-arid)*
- *BWk (Cold desert)*
- *BWh (Hot desert)*
- *Aw (Savanna)*
- *Am (Monsoon)*
- *Af (Rainforest)*

**Isotherm used to separate temperate (C) and continental (D) climates is -3°C*
Data source: Climate types calculated from data from WorldClim.org

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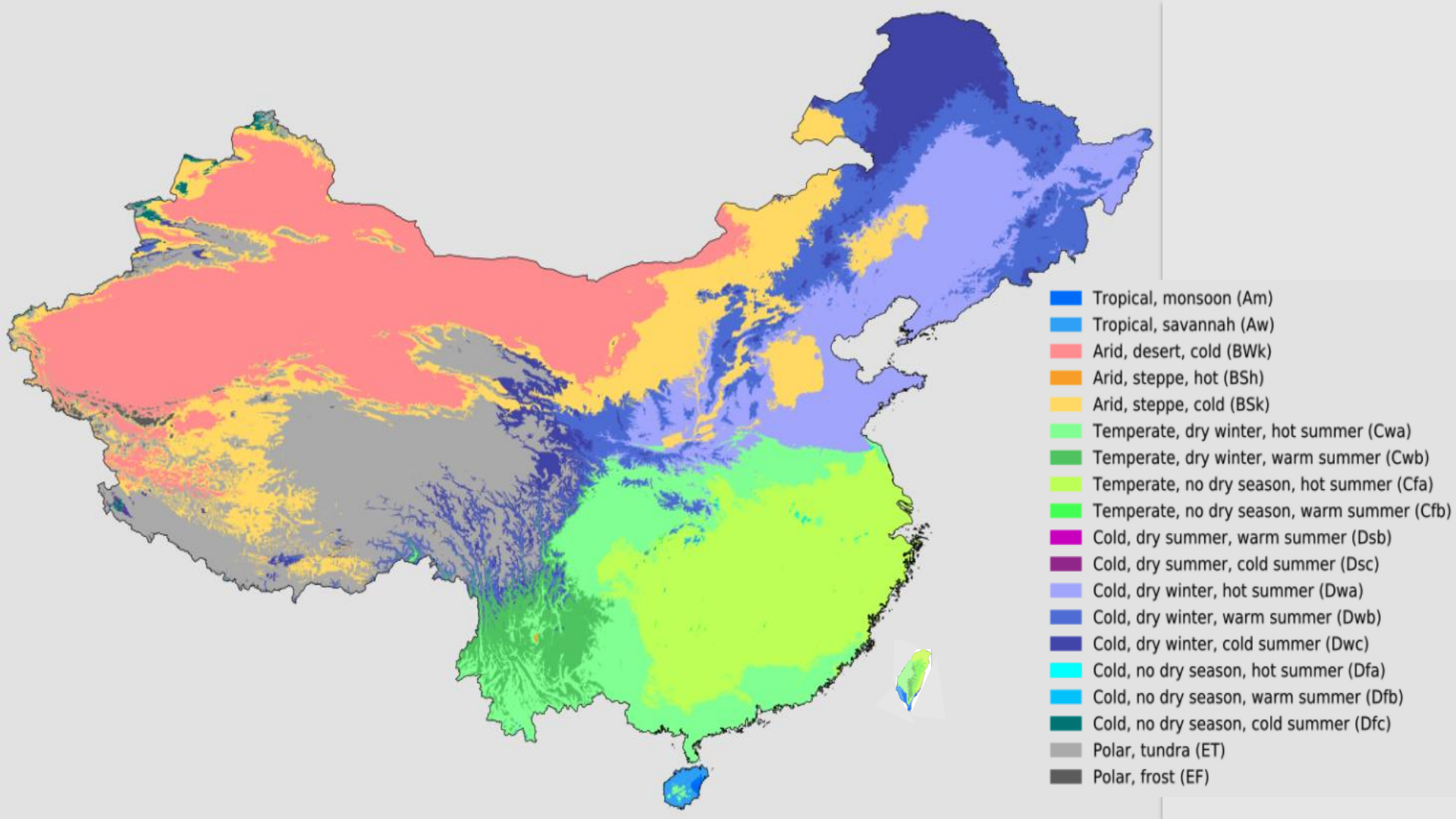
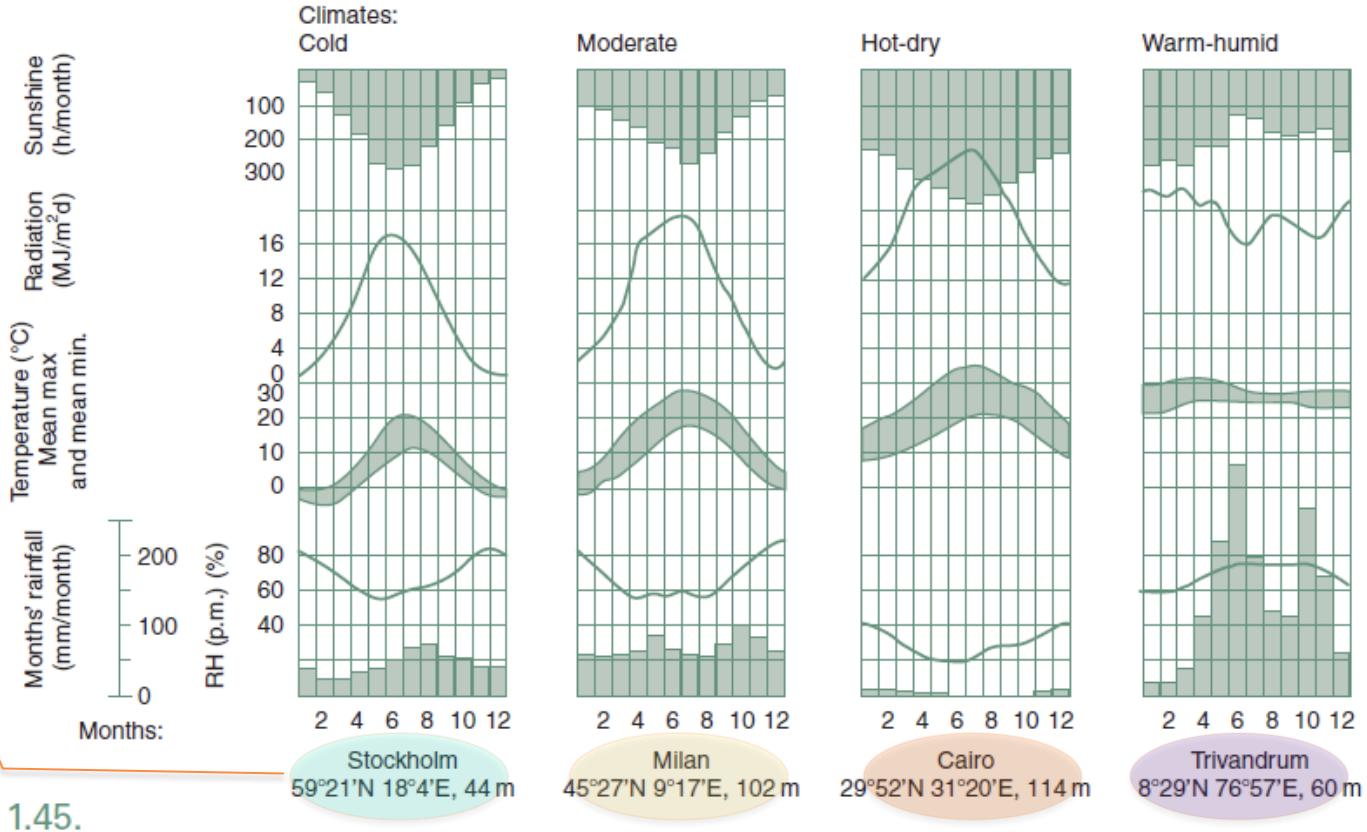

Tropical, monsoon (Am)
Tropical, savannah (Aw)
Arid, desert, cold (BWk)
Arid, steppe, hot (BSh)
Arid, steppe, cold (BSk)
Temperate, dry winter, hot summer (Cwa)
Temperate, dry winter, warm summer (Cwb)
Temperate, no dry season, hot summer (Cfa)
Temperate, no dry season, warm summer (Cfb)
Cold, dry summer, warm summer (Dsb)
Cold, dry summer, cold summer (Dsc)
Cold, dry winter, hot summer (Dwa)
Cold, dry winter, warm summer (Dwb)
Cold, dry winter, cold summer (Dwc)
Cold, no dry season, hot summer (Dfa)
Cold, no dry season, warm summer (Dfb)
Cold, no dry season, cold summer (Dfc)
Polar, tundra (ET)
Polar, frost (EF)

Climate

Building design factors

Typical cities



1.45.

Composite (simplified) climate graphs for the four basic types.

Source: Szokolay (2008: 35)

Climate types

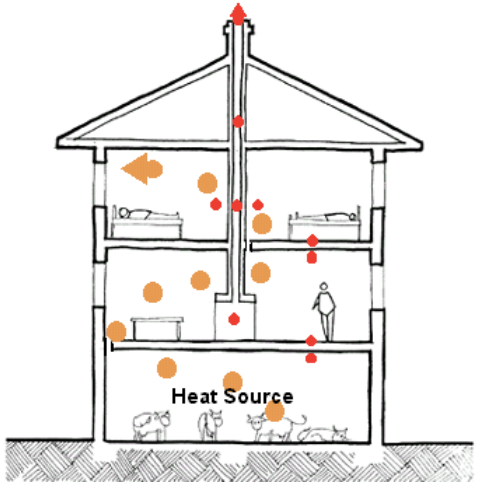
Cold climates

- ✓ Lack of heat (from the ground), or an excessive heat dissipation for all or most of the year

Climate types

Cold climates

- ✓ Surface/volume ratios
- ✓ Insulation
- ✓ Sheltered entrances
- ✓ Infiltration/exfiltration
- ✓ Small windows
- ✓ Minimal building mass



Climate types

Temperate climates

- ✓ Surface/volume relaxed
- ✓ Insulation optimized
- ✓ Solar heat desirable for winter
- ✓ Building mass
- ✓ Ventilation
- ✓ Windows
- ✓ Deciduous vegetation

Climate types

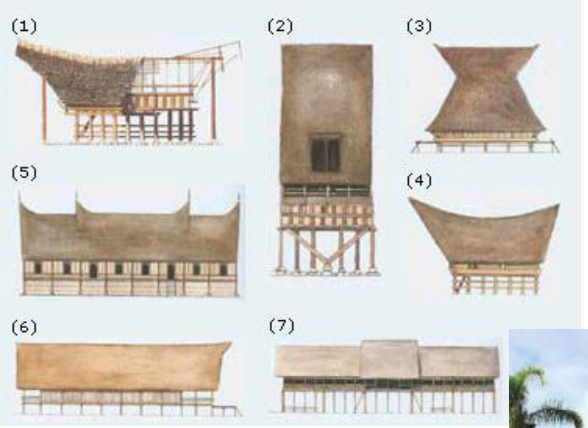
Hot-dry climates

- ✓ Protection from hostile outdoors
- ✓ Control solar gain
- ✓ Reflective colours
- ✓ Windows: small to admit little sun
- ✓ Large thermal mass for daytime quarters, but separate sleeping areas
- ✓ Ventilation during the day minimal
- ✓ Evaporative cooling
- ✓ Vegetation

Climate types

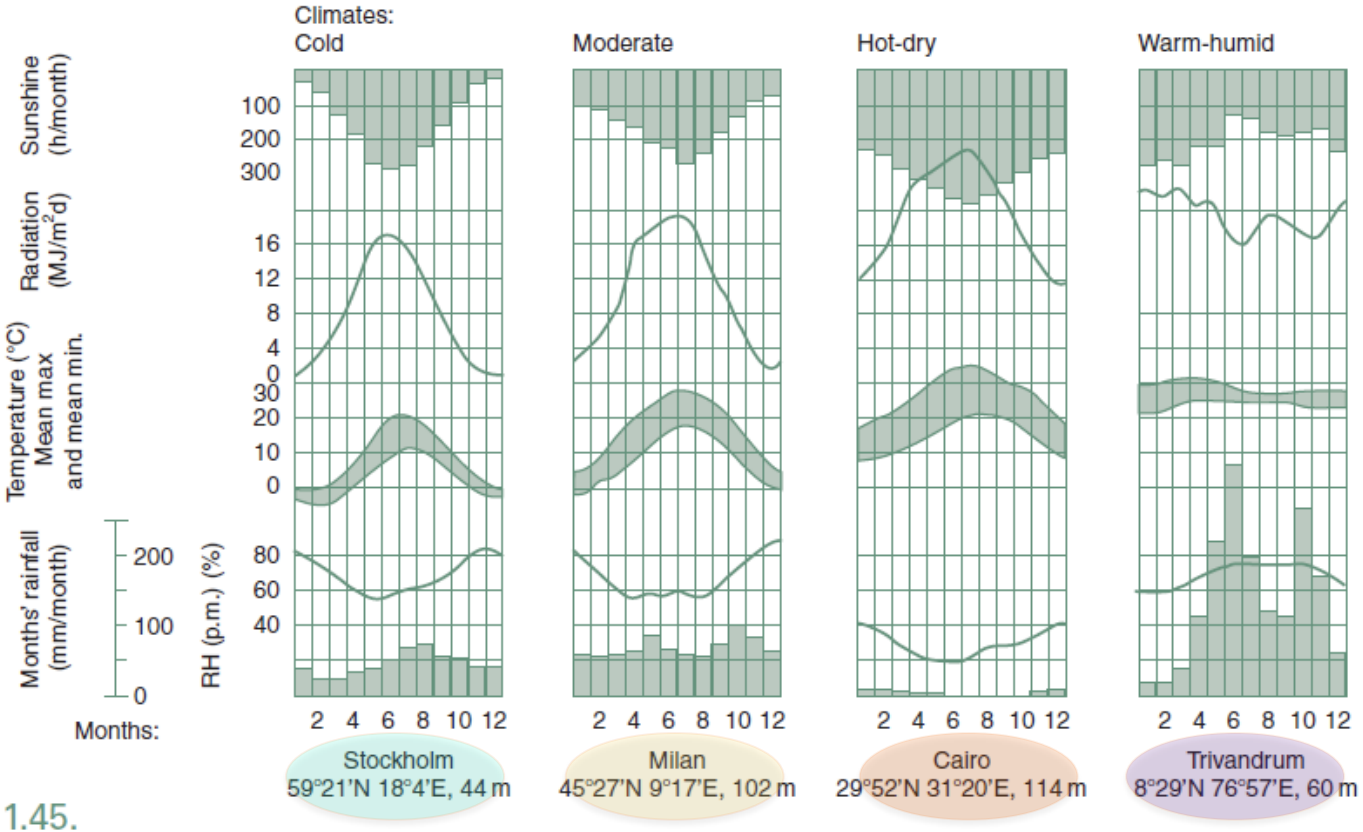
Warm-humid climates

- ✓ Overheating is not as great as in hot-dry areas
- ✓ High humidity restricts the evaporation potential
- ✓ The diurnal temperature variation is small (difference between night and day temperature



Climate

Building design factors



1.45.

Composite (simplified) climate graphs for the four basic types.

Source: Szokolay (2008: 35)

Climate

Building design factors

- Discuss some *limitations* of these climate assumptions?

HOMEWORK:

For your hometown and XuZhou, find:

1. the longitude and latitude of both cities,
2. the Koppen-Geiger classification of both cities (e.g. Cfa, etc), and
3. Describe the weather profile of both cities (temperature range, rainfall, sunshine, radiation)

Building energy use

… and energy conservation

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Building energy use

Reducing Energy use?

- ✓ **Energy conservation** measures to minimise the supplementary heating and/or cooling needs
  - ✓ (e.g. Wall, floor and roof insulation, double or triple glazing, etc)

- ✓ This is also called '**reducing the DEMAND'** (for electricity consumption) that a building needs to operate.

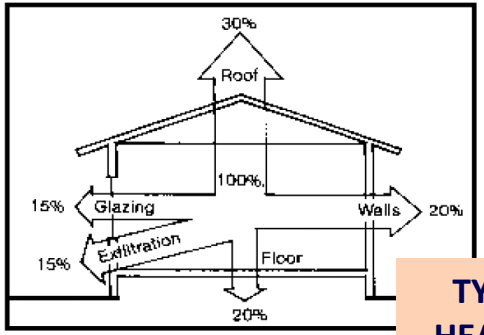
- ✓ **Climate responsive design** – making the most of Nature and passive cooling and heating techniques.

Building energy use

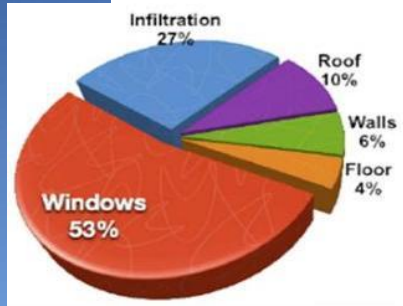
Energy conservation

- ✓ The amount of energy used for cooling or heating in buildings MUST be **reduced** through good design
- ✓ Energy losses vary depending on the building type

We must control heat movement to make our buildings more comfortable!

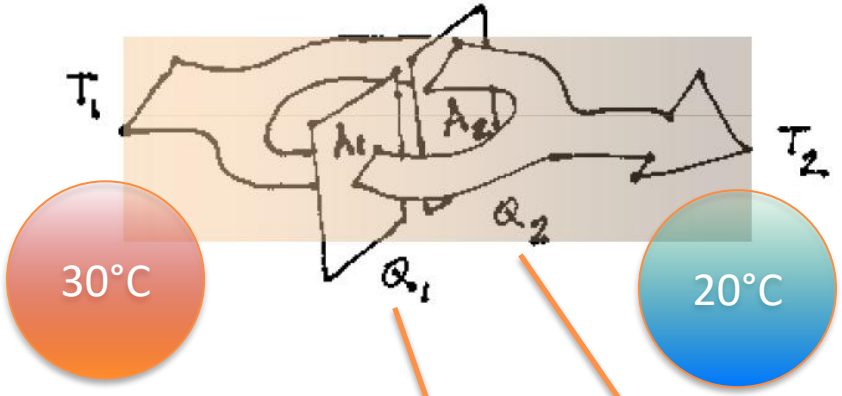


TYPICAL HEAT LOSS SOURCES



Building energy use

Total conductive heat flow



$$Q_{total} = Q1 + Q2$$

✓ Parallel heat flow – heat moves through all surfaces of a building

Building energy use

Every modern building has:

- heating and/or cooling **LOAD,** and heating and/or cooling **DEMAND**

✓ Annual heating DEMAND, Q_H (kWh/year)

$$Q_H = (Q_t + Q_v) - \eta \times (Q_s + Q_i)$$

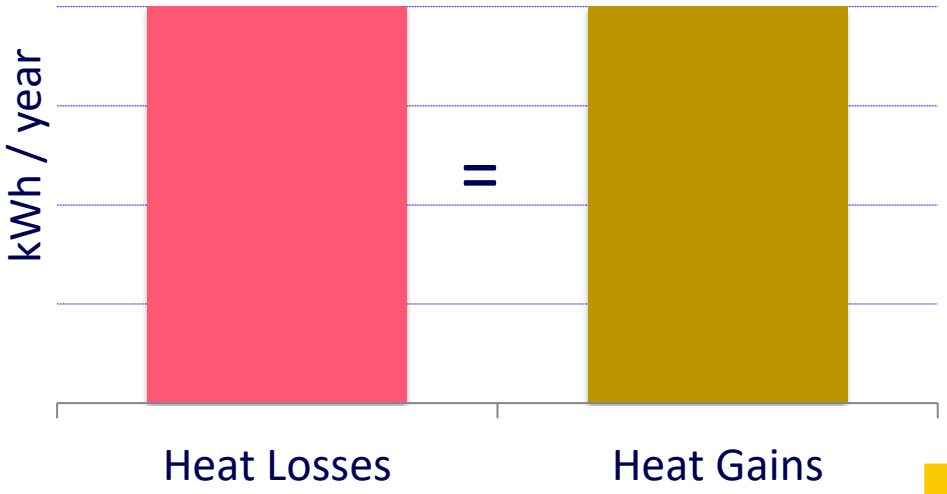
Where:

- Q_H = Annual space heat demand (energy required for heating)
- Q_t = Transmission heat losses (energy lost through walls, windows, roof, etc.)
- Q_v = Ventilation heat losses (energy lost through air exchange, including infiltration and mechanical ventilation)
- η = Utilization factor for free heat gains (the efficiency of using available free heat gains)
- Q_s = Solar gains (energy gained from sunlight)
- Q_i = Internal heat gains (energy gained from internal sources like appliances, lighting, and occupants)

Building energy use

Energy balance:

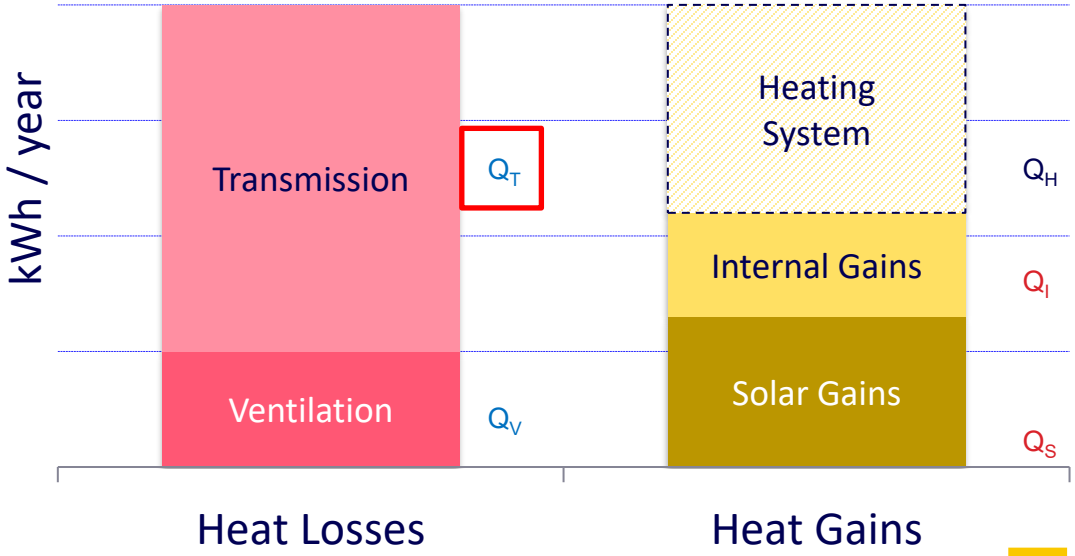
- The heat losses must be balanced by heat gains to the space to maintain a constant temperature.



Building energy use

$$Q_H = (Q_t + Q_v) - \eta \times (Q_s + Q_i)$$

Energy balance



Building energy use

U-value concept

$$U = \frac{1}{R}$$

- ✓ U-value is the reverse thermal resistance of a building element (wall, roof, window, etc)
- ✓ U-value $\left(\frac{W}{(m^2.K}\right)$
- ✓ R-value (= U-value^{-1})

U = W/(m^2.K)
W = Watts
K = Kelvin

✓ Advantages of the U-value concept

- ✓ Fast to calculate
- ✓ Allows optimisation of single parts

✓ Disadvantages of the U-value concept

- ✓ Not exact
- ✓ Only applicable for flat surfaces (e.g. walls)
- ✓ Doesn't allow for connections between different materials

Building energy use

$$U = \frac{1}{R}$$

U-value concept

$$U = \frac{1}{R_T} = \frac{1}{R_{si} + \frac{d_1}{\lambda_1} + \frac{d_1}{\lambda_1} + \frac{d_1}{\lambda_1} + R_{se}}$$

- ✓ U-value is the reverse thermal resistance of a building element (wall, roof, window, etc)
- ✓ U-value $\left(\frac{W}{(m^2.K)}\right)$
- ✓ U-value is the thermal TRANSMITTANCE
- ✓ R, R-value is the thermal RESISTANCE

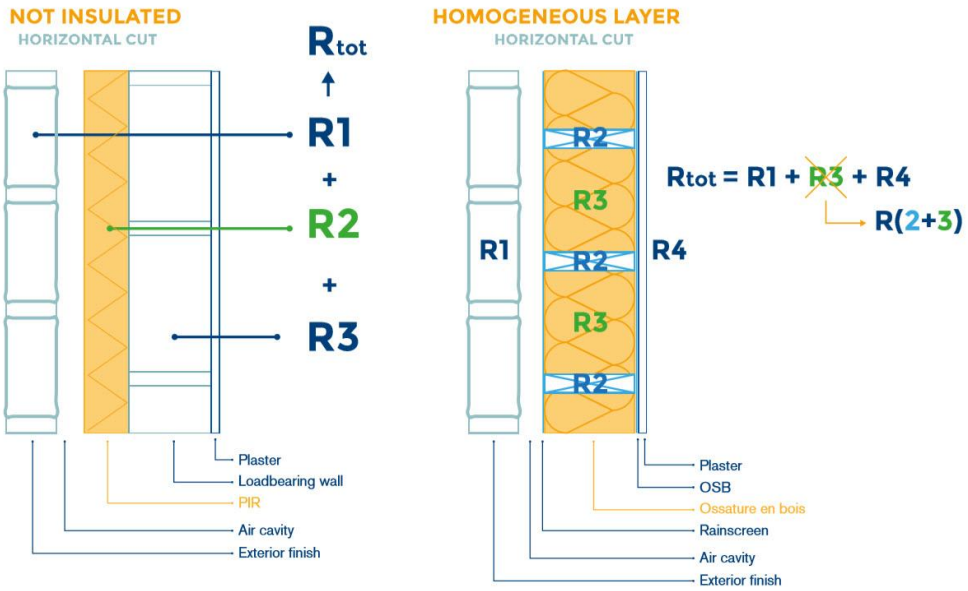
- ✓ R_T : Total thermal resistance
- ✓ d_n : thickness of component layer (m)
- ✓ λ_n : thermal conductivity (W/m.K)
- ✓ R_{si} : Heat transfer resistance at interior surface
- ✓ R_{se} : Heat transfer resistance at exterior surface

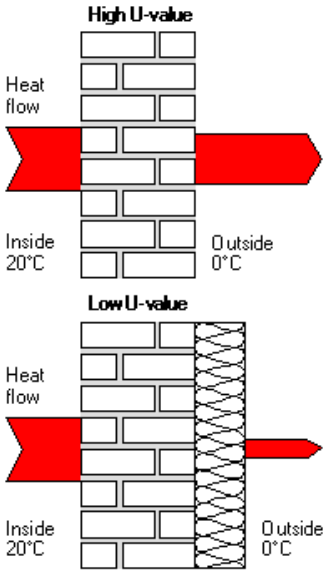
Building energy use

$$U = \frac{1}{R}$$

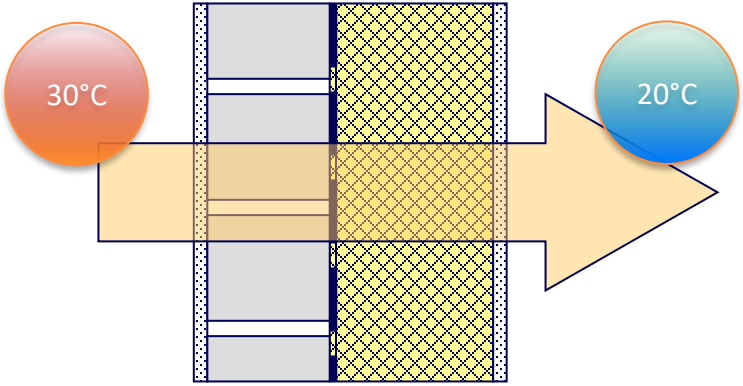
$$U = \frac{1}{R_T} = \frac{1}{R_{si} + \frac{d_1}{\lambda_1} + \frac{d_1}{\lambda_1} + \frac{d_1}{\lambda_1} + R_{se}}$$

U-value concept





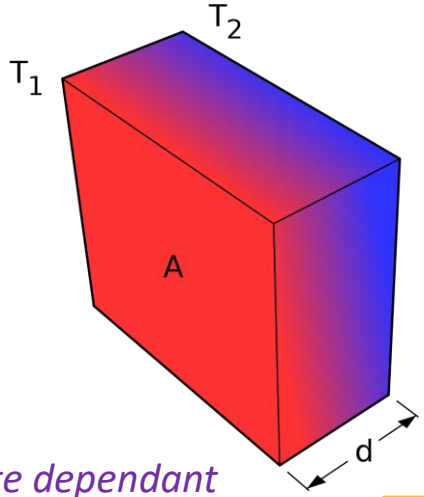
Building energy use



Total conductive heat flow

- ✓ Parallel heat flow – heat moves through all surfaces of a building – but a different rates!

$$Q_T = A * U * b_t * G_t$$



- ✓ Q_T: Transmission heat loss ($kWh/year$)
- ✓ A : Area of wall (m^2)
- ✓ U : Heat transfer coefficient (also called the 'U-value') ($\frac{W}{(m^2.K}$)
- ✓ b_t : Temperature correction factor (if exposed to outside = 1)
- ✓ G_t : Heating degree hours = time integral of temperature differences between outside and inside ($k.Kh/year$) – *this is climate dependant*

Building energy use

Heating degree-days, G_t

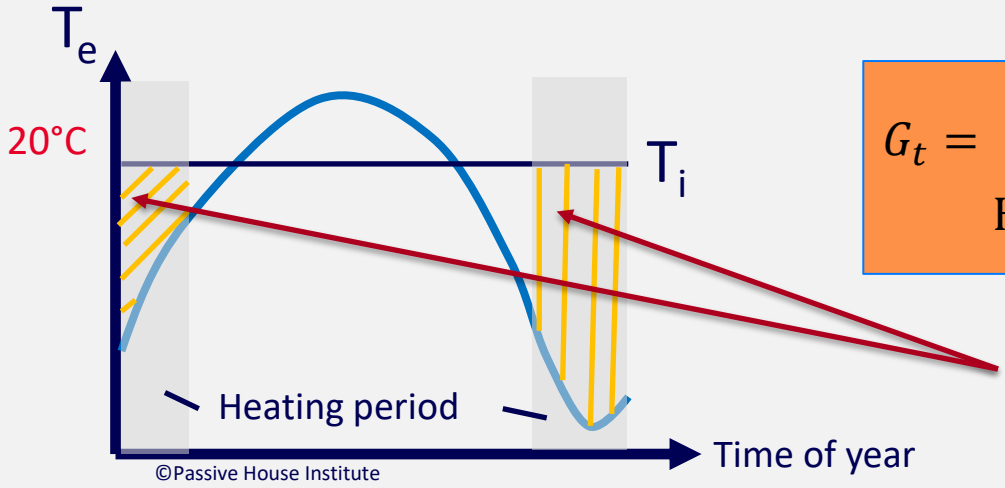
(also known as ‘HDD’, also measured in hours)

- ✓ The number of days (and hours) that heating is required in your building
- ✓ Historically the base *internal* temperature is set at 20°C
- ✓ If the *outside* temperature is > 20°C temperature ⟹ No heating

Building energy use

Heating degree-days, G_t

G_t is the sum of the temperature differences between the inside and the outside air temperature during heating period



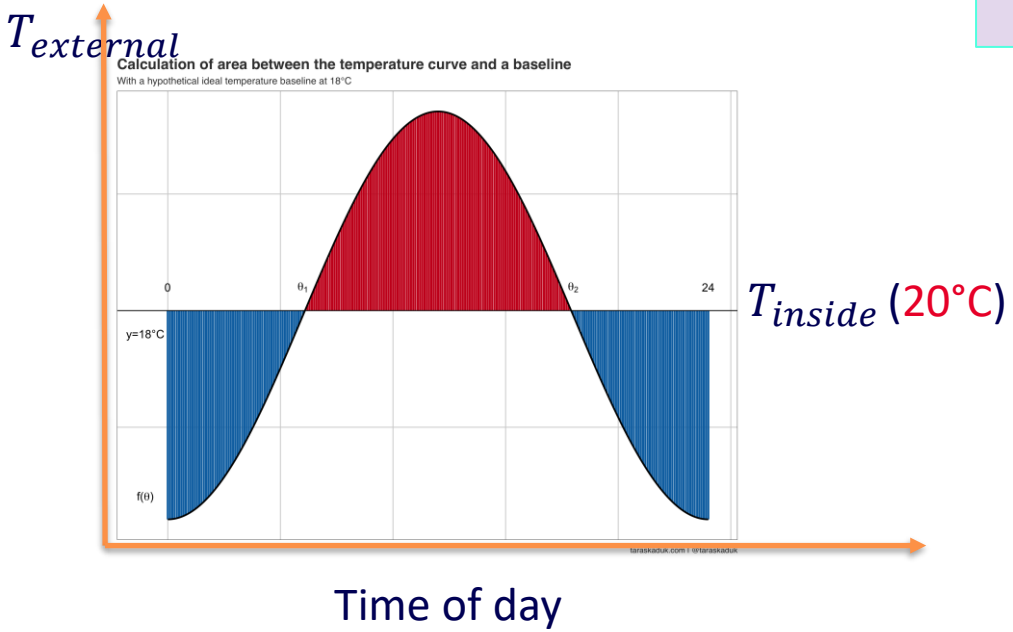
$$G_t = \int_{\text{Heating period}} (T_i - T_e).\,dt$$

Heating period (Northern hemisphere)

Building energy use

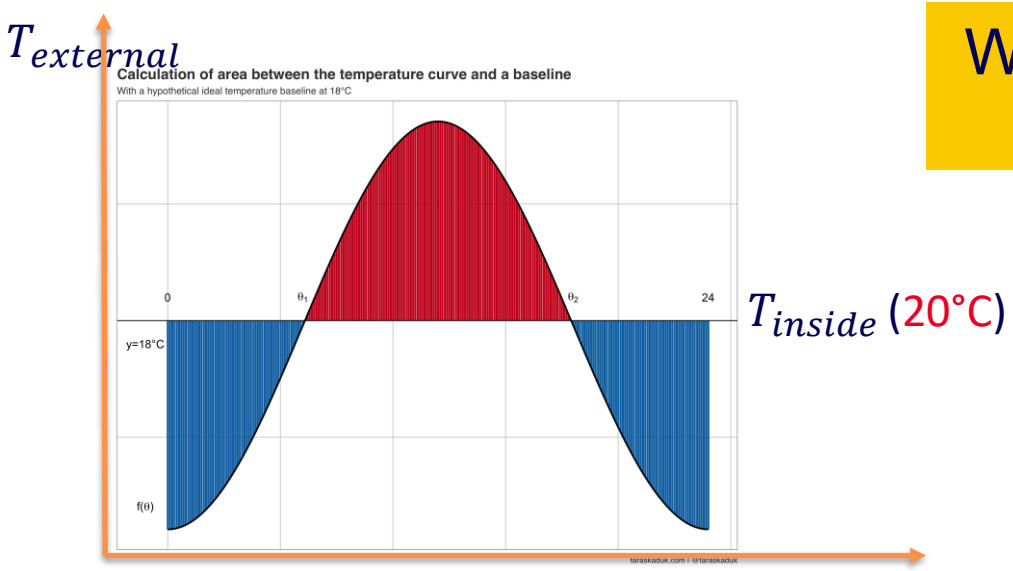
Heating degree-hours, G_t

$$G_t = \int (T_{inside} - T_{external}).dt$$



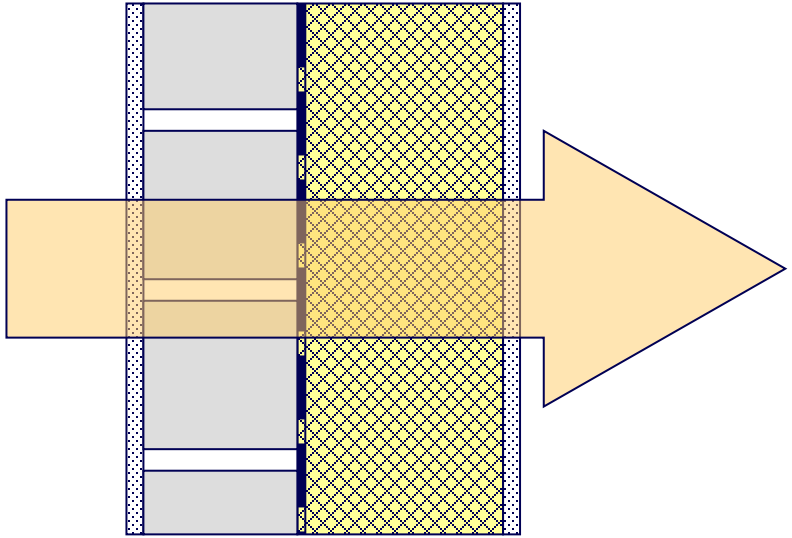
Building energy use

Heating degree-hours, G_t



What is the G_t for XuZhou?

Building energy use



Example

$$Q_T = A * U * b_t * G_t$$

- ✓ Q_T: ??? ($kWh/year$)
- ✓ A : Area of class room walls (m^2)
- ✓ U : 0.5 ($\frac{W}{(m^2.K}$)
- ✓ b_t : 1
- ✓ G_t : 100 ($k.Kh/year$) – Just a guess for XuZhou

Area of thermal envelope * U-value * Temperature-correction factor * Heating degree hours

1 m² * 0.5 W/(m²K) * 1.0 * 100 kKh/a = __50__ kWh/a

____ m² * 0.5 W/(m²K) * 1.0 * 100 kKh/a = _____ kWh/a

Building energy use

Cooling degree-days ('CDD')

- ✓ Conceptually similar to the heating degree-days
- ✓ The base temperature is usually taken as 25 to 26∘C

$$CDD = \Sigma(T_{av} - 26)$$

From day 1 to 365

mean temperature of the day (T_{av})

Building orientation

… and energy conservation

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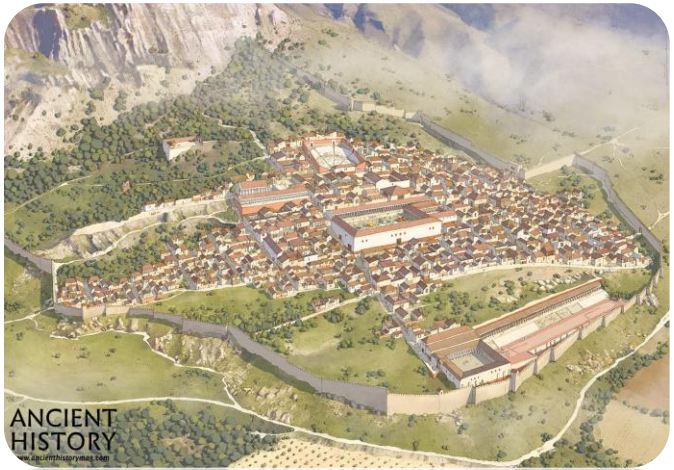
Building orientation

Ancient examples



Priene, Turkey 350 B.C.

"In winter, rays from the sun traveling low across the southern sky streamed across the south-facing courts, through the portico, and into the house - heating the main rooms."

- 'Earthship' style home

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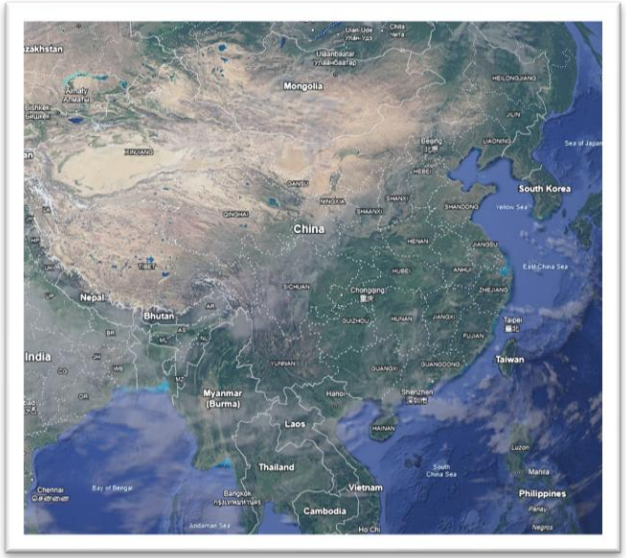
Building orientation

Ancient examples



China, ~1500 B.C.

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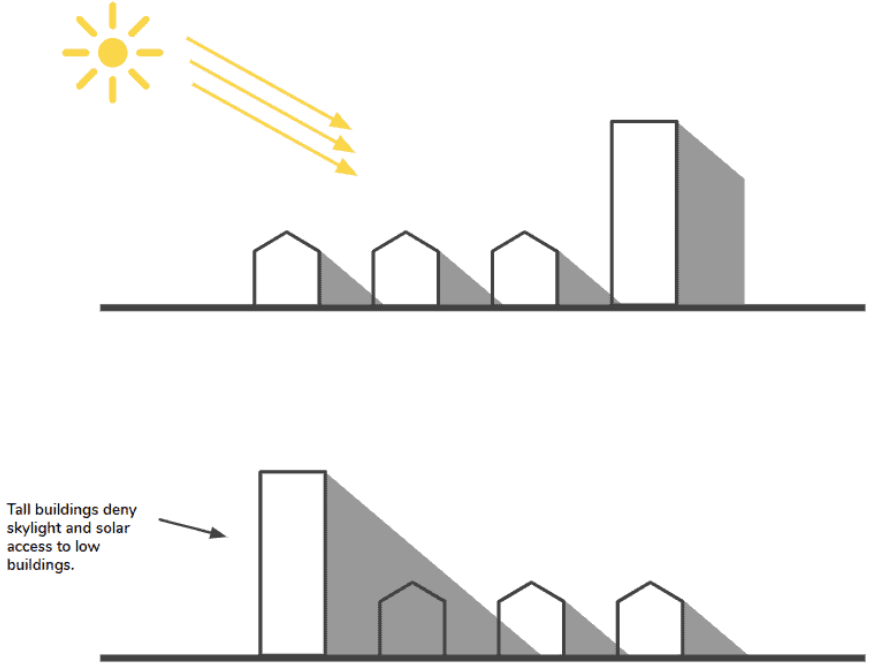
Source: Solar Architecture in Ancient China

Building orientation

Solar gain

- ✓ Sun provides radiant heat
- ✓ Buildings benefit from this ‘free’ source of energy
- ✓ Building orientation and position is very important
- ✓ Window location is also very important



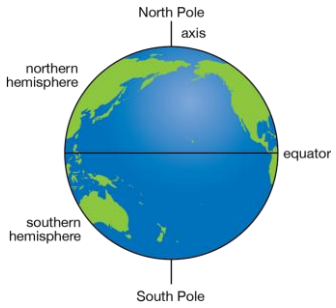
Source: https://planlux.net/planning-for-solar-access-building-orientation/

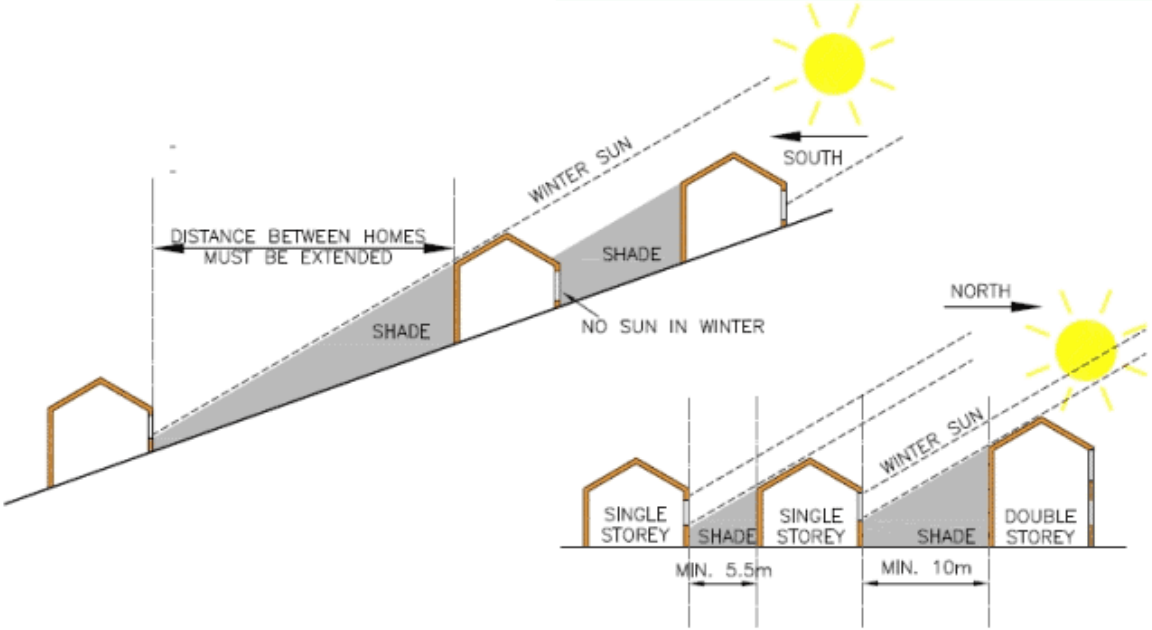
Building orientation

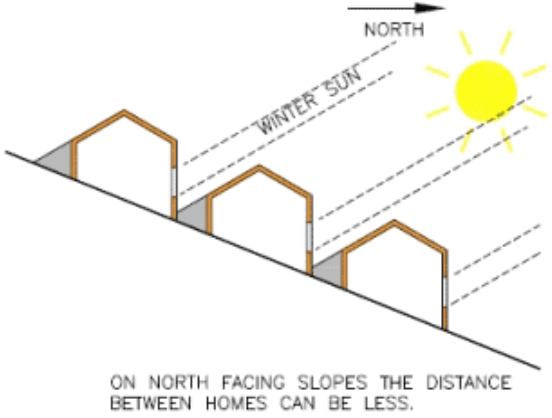
Solar gain

Northern hemisphere = South facing for maximum solar gain

Southern hemisphere = North facing for maximum solar gain





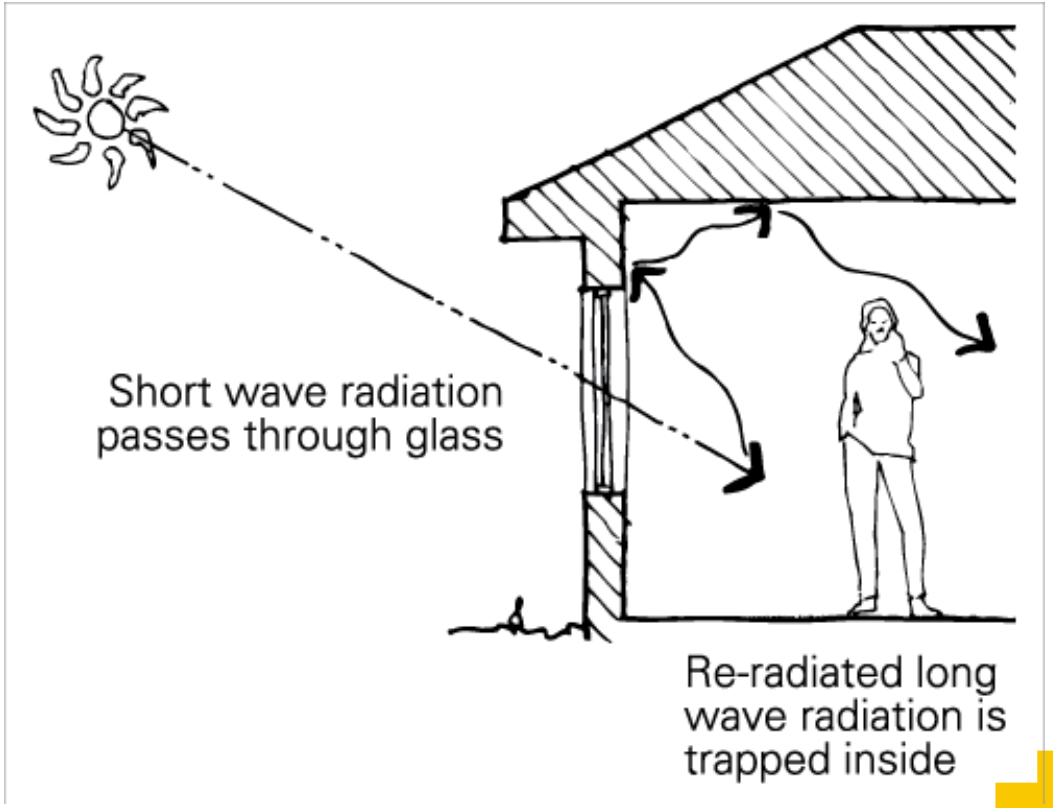


Source: https://planlux.net/planning-for-solar-access-building-orientation/

Building orientation

Solar gain

- ✓ Glass helps to trap solar radiation
- ✓ ‘Glass house’ effect

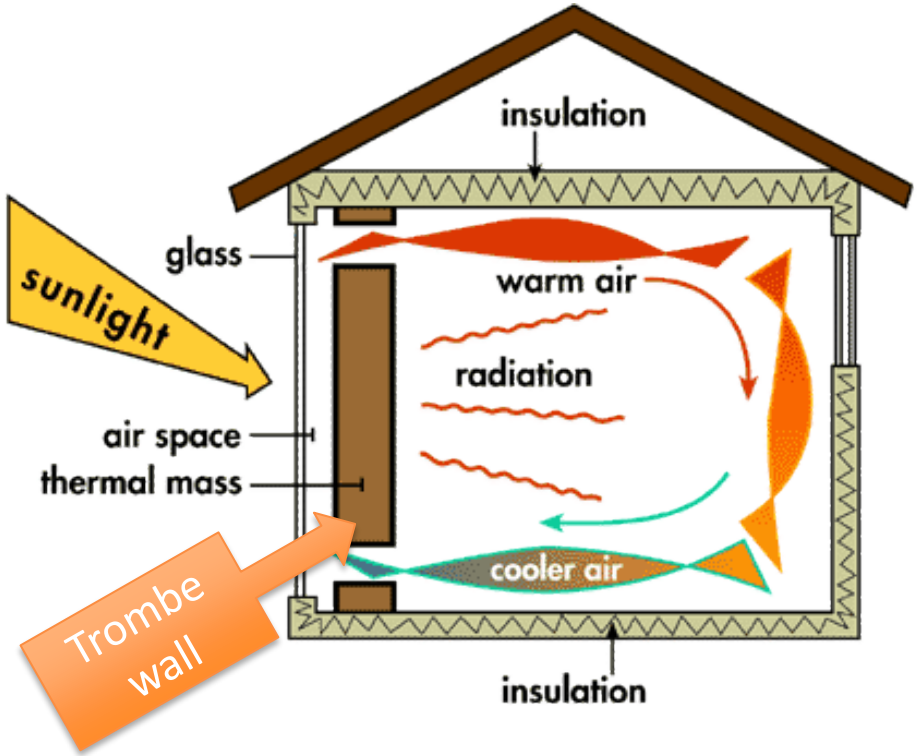


Climate strategies

Passive solar heating

- ✓ Ways to collect solar energy
- ✓ Ways to keep heat inside the building
- ✓ Ways to store heat from the sunny part of the day, until it is needed later in the evening

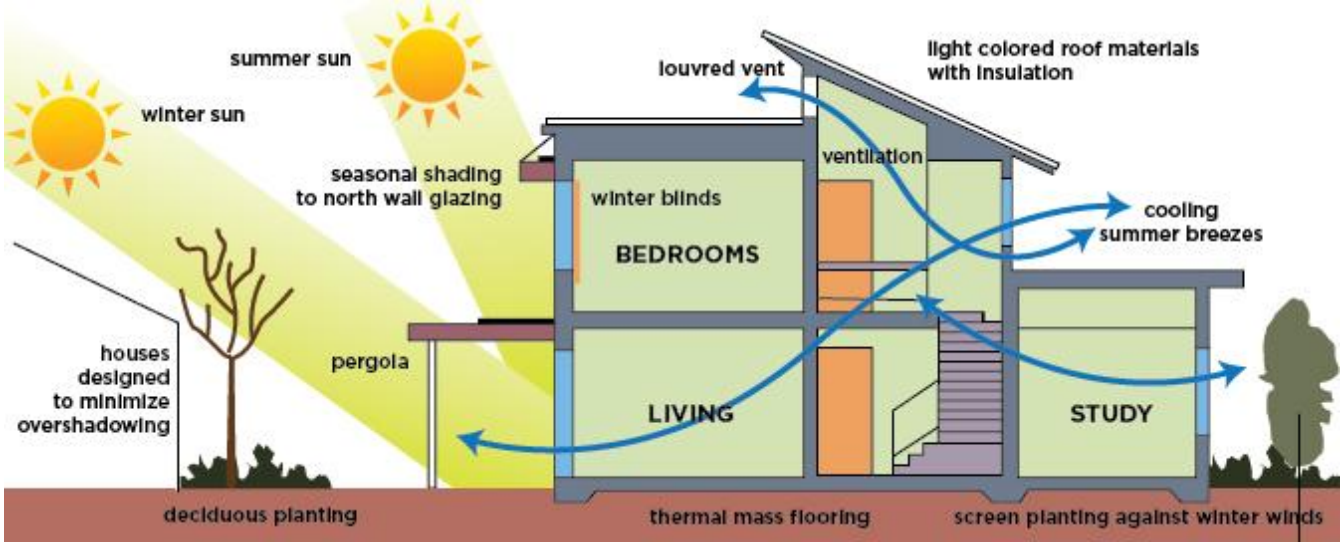
(e.g. black carpet or floors, Trombe walls)



Building orientation

Summer and Winter

- ✓ Reduce the solar gain in Summer
- ✓ Increase solar gain in Winter
- ✓ Deciduous plants – shade in Summer, solar gain in Winter

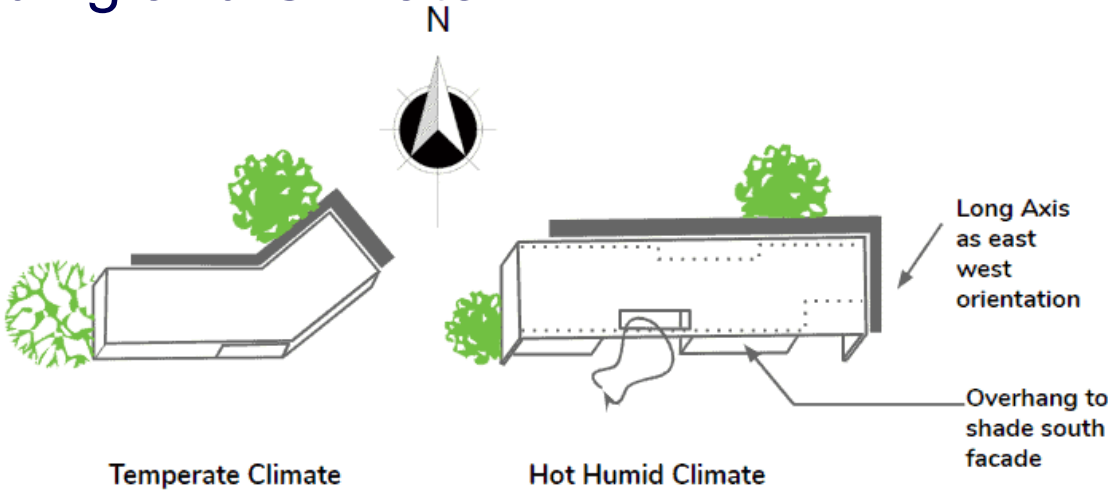


Source: https://planlux.net/planning-for-solar-access-building-orientation/

Building orientation

Building shape, shading and Climate

- ✓ Building orientation in different climates is important
- ✓ External shading prevents excess solar gain in Summer
- ✓ Minimising East and West walls of building helps reduce energy use



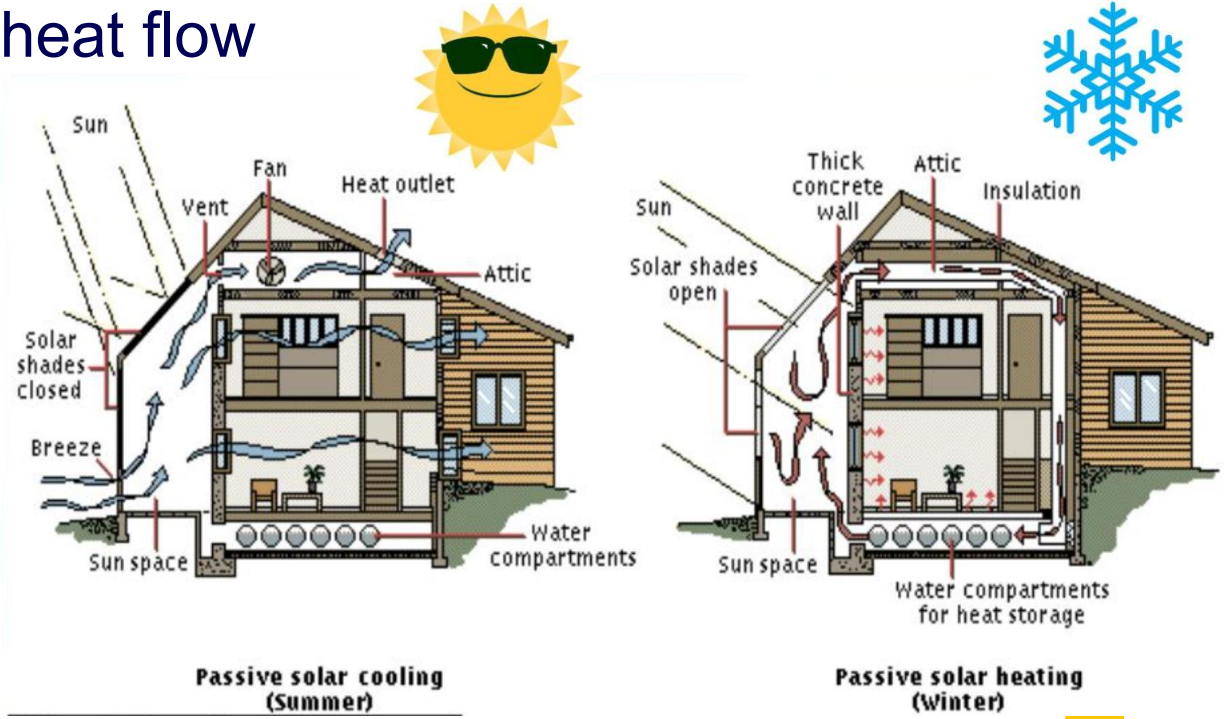
Southern hemisphere example

Source: https://planlux.net/planning-for-solar-access-building-orientation/

Climate strategies

Passive control of heat flow

- ✓ Passive solar heating
- ✓ Passive cooling
- ✓ Thermal mass
- ✓ Natural ventilation



Thank you

Next time: PASSIVE CONTROL STRATEGIES

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